

Project Workflow: Week 4

Phase: Game Window & Core Loop Setup

1. Objective

The goal of Week 4 was to bring the game window to life by establishing the core update cycle. This included:

1. Implementing the GamePanel class with the core update/repaint cycle.
2. Setting up a javax.swing.Timer to drive a ~60 FPS game loop.
3. Establishing the 900x700 game window.

2. Detailed Task Breakdown

▪ Implementing the GamePanel Update/Repaint Cycle

Built out the GamePanel class (JPanel) so it can drive the game's core cycle: an update() method to advance game state each frame, and a paintComponent() override to render the current state to the screen. This separates game-state updates from rendering, which keeps the panel ready to hold the Road, PlayerCar, EnemyCar, Environment, and GasItem objects designed in Week 2 as they are implemented in later weeks.

▪ Setting Up the Timer-Based Game Loop

Configured a javax.swing.Timer to fire at roughly 60 frames per second (approximately every 16-17 milliseconds), calling update() and then repaint() on each tick. Using Swing's own Timer keeps every update safely on the Event Dispatch Thread, avoiding the threading issues that come with a manual loop or a separate thread driving UI updates.

▪ Establishing the Game Window

Fixed the game window at 900x700 pixels, matching the Week 2 wireframe, and connected GamePanel to the CarRacingGame JFrame created in Week 3. Set up paintComponent(), including enabling anti-aliasing via Graphics2D, so that later weeks can layer in the road, scenery, cars, and HUD without having to revisit the rendering setup.

3. Deliverables

1. GamePanel Class: A working JPanel with a defined update/repaint cycle.

2. Game Loop: A `javax.swing.Timer` running at ~60 FPS, driving `update()` and `repaint()` each tick.
3. Game Window: A working 900x700 window with `GamePanel` attached.
4. Base Rendering Pipeline: A `paintComponent()` setup ready to support future rendering layers.

4. Tools & Resources

- Development Environment

1. Java JDK 8+
2. IntelliJ IDEA / VS Code

- Reference Materials

Java Swing Timer documentation, `Graphics2D` and `paintComponent()` rendering references, and `javax.swing.Timer`-based game loop tutorials.